

Can intercrops improve soil water infiltrability and preferential flow in rubber-based agroforestry system?

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ABSTRACT

Land-use change due to the widespread practice of rubber-based agroforestry influences ecosystem services. However, little is known about their soil hydrological functioning associated with infiltration and water flow behaviour which affects surface runoff, soil erosion, and groundwater recharge. In order to clarify whether the intercrop can promote soil water infiltrability and water movement in rubber-based agroforestry system, 66 field infiltration and 2 dye tracer experiments were performed in this study on undisturbed soil in a rubber-tea agroforestry system in Xishuangbanna, SW China. The results showed that the field saturated soil hydraulic conductivity (K_s) exhibited a high spatial heterogeneity. The K_s , initial infiltration rate (I_s), and actual steady-state infiltration rate (I_s) were significantly higher in the tea tree planting zone (TT) than in the rubber tree planting zone (RT). The K_s decreased with distance from the trunks of tea trees, and the K_s between rubber trees in a row was lower than that between rubber rows. Along with improved soil physical properties, increased root and faunal activities, and more inhomogeneous cracks and stones, there was higher soil water infiltrability and more preferential flow in the TT than in the RT. Therefore, tea shrubs in rubber agroforestry systems may function as water harvesters, contributing to deeper drainage and recharge. In contrast, the rubber trees function as water consumers, causing soil desiccation and water scarcity. Differences in the soil hydraulic properties among different zones in a rubber-tea agroforestry system can lead to the spatial redistribution of surface and belowground water, increasing water availability for various plants with different root systems. Better infiltration and increased preferential flow beneath tea trees potentially reduce runoff generation and erosion risk, promote groundwater recharge, and increase water storage that may counteract the interception and transpiration losses from intercrops and rubber trees, thereby contributing to the management of water resources.

1. Introduction

Rubber plantations are rapidly expanding throughout mainland Southeast Asia (Xu, 2011). By 2050, the area of land dedicated to rubber is predicted to more than double or triple (Fox et al., 2014; Ziegler et al., 2009). In the Yunnan Province in China, rubber monoculture expanded to cover 22% of the Xishuangbanna landscape (Xu et al., 2014). The conversion of primary tropical forests to rubber has resulted in several environmental and ecological issues, such as reduced biodiversity (Warren-Thomas et al., 2015), lower carbon stocks, hard and crusted soil (Liu et al., 2015), reduced rainwater infiltration (Ziegler et al., 2009), excessive water loss and soil erosion (Liu et al.,

2016), negative hydrological consequences (Fox et al., 2014), and potentially impaired soil- and groundwater recharge. In response to such problems, constructing rubber-based agroforestry systems were proposed by the Chinese government and scientists (Feng, 2007). In these agroforestry systems, rubber trees are commonly intercropped with fruits, economic plants, and traditional medicinal plants. For example, approximately 5% of the rubber monoculture has been replaced by rubber-tea agroforestry systems (Liu et al., 2016). Previous studies highlighted that rubber-based agroforestry systems (e.g., rubber with tea and cacao) achieved a desirable water use strategy (Wu et al., 2016) and could substantially reduce water and soil losses by improving vegetation traits and soil management practices (Chen et al., 2017; Liu

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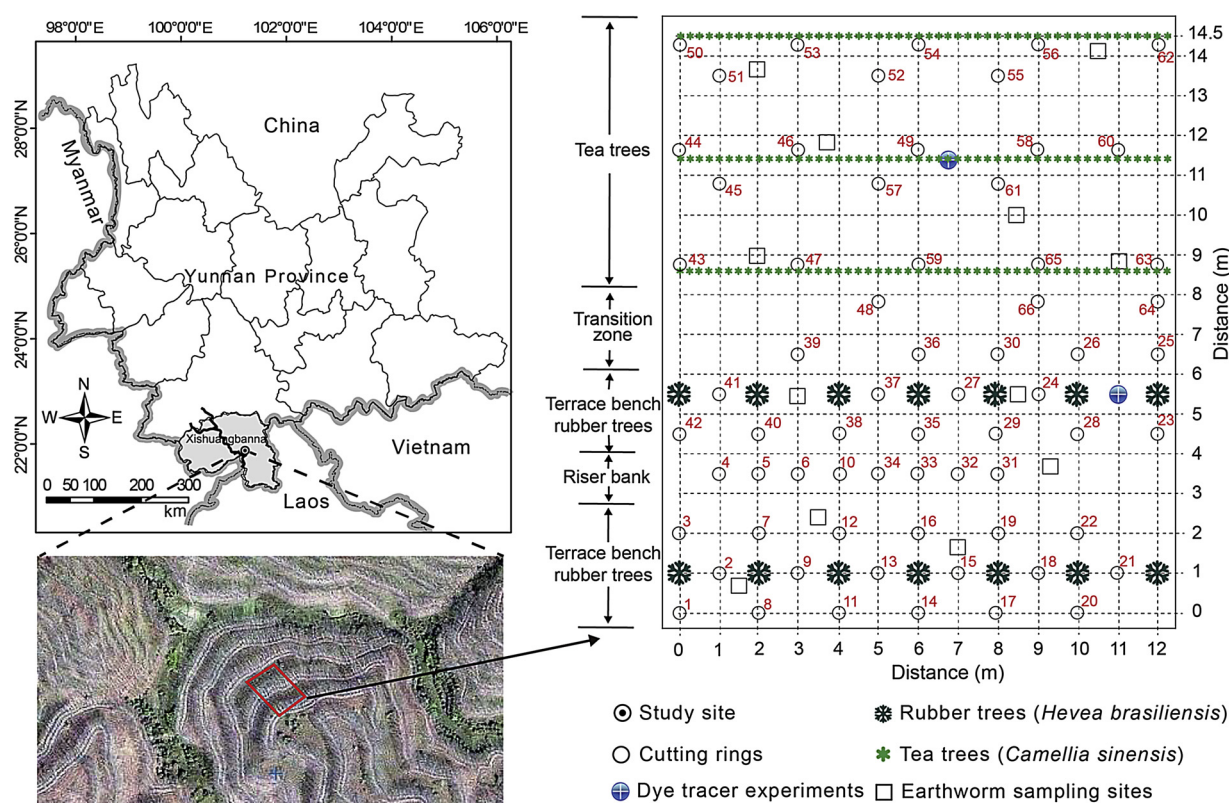


Fig. 1. Location of the study site in Xishuangbanna, SW China. Maps show the spatial distribution of field measurements and experimental design of the study plot.

et al., 2016; Zhu et al., 2018). Although such systems are considered to be economically viable and ecologically sustainable (Snoeck et al., 2013), our understanding of how their hydrological ecosystems services (e.g., infiltration and storage of water, and regulation of water supplies) are affected by intercropping practice is rather limited (Benegas et al., 2014; Ziegler et al., 2009; Liu et al., 2016).

Trees can modify microclimate and hydrological processes that affect the water balance, such as transpiration, runoff generation, infiltration, groundwater recharge, preferential flow, lateral flow, and water exchange (Charbonnier et al., 2013; Bargués Tobella et al., 2014; Benegas et al., 2014). In the rubber-based agroforestry systems, the water balance is closely related to the growth and survival of the plant community. Rubber trees have high water consumption rates due to their large vessels, exerting negative effects on water resources (Ayutthaya et al., 2011). Results reported by Tan et al. (2011) show that the high evapotranspiration rates of rubber trees resulted in zero flow and water scarcity during the dry season; decades of rubber cultivation caused low streamflow and dried wells in many villages (Qiu, 2009). Water availability associated with plant growth is inevitably influenced. Although a recent study found that rubber trees in agroforestry systems have strong plasticity of water uptake and capacity to avoid interspecific competition for water (Wu et al., 2016), water absorption often changes with different root distribution patterns (Liu et al., 2014). Rubber trees have well-developed root systems with 1.5–2.4 m long taproots and 6–9 m long laterals (Priyadarshan, 2011), whereas intercropped perennial shrubs (e.g., tea trees and coffee) have short lateral roots and a moderate number of fine roots that overlap with the roots of the rubber trees in the shallow soil layer (Wu et al., 2016). Macropores formed by the roots' growth and decay, coupled with earthworm burrows, termite holes, and structural cracks and fissures can provide preferential flow pathways for water flow into different soil layers, which contributes to the redistribution of soil water (Bromley et al., 1997; Devitt and Smith, 2002). Abundant lateral roots promote the lateral water movement. Water flows into the surrounding

soil matrix through macroporous walls with the help of capillary action, enhancing the water exchange (Wine et al., 2012). Dye tracer experiments are a valuable technique for characterizing preferential flow (Alaoui et al., 2011; Flury and Flühler, 1995). By processing the original dye-stained soil profiles through image analysis procedures, dye patterns could be classified and different types of water flow behaviour (e.g., preferential flow, lateral flow and water exchange) and their quantitative information can be acquired from classified dye patterns (Cey and Rudolph, 2009; Jiang et al., 2018).

Water infiltration into soil is generally affected by management practice, vegetation cover (Castellano and Valone, 2007), soil faunal activities (Alaoui et al., 2011; Bargués Tobella et al., 2014), and compaction by machine traffic and trampling livestock (Wang et al., 2015). In rubber agroforestry systems, daily latex tapping and herbicide application may influence water infiltration by affecting soil properties of certain location. Moreover, intercropped species often produce more residues and thus improve soil porosity and organic matter content and facilitate the water infiltration and storage (Ilstedt et al., 2007). The saturated hydraulic conductivity (K_s), a key soil hydraulic parameter, plays a crucial role in modelling the water movement and transport processes of dissolved solutes through the soil profile (Jačka et al., 2014; Reynolds et al., 2000). A high K_s can enhance rainfall infiltration and reduce surface erosion. Most studies support the idea that trees enhance K_s (Burch et al., 1987; Chandler and Chappell, 2008; Lorimer and Douglas, 1995), however, in only few studies the effects of intercropping on the K_s in the rubber-based agroforestry systems were investigated.

To determine the hydrological functions of rubber agroforestry systems and the role of intercropped species, we selected a rubber-tea agroecosystem and investigated the variation of soil infiltrability and water flow behaviour. Specifically, the aims of this study were to: (1) investigate the spatial variability of infiltrability by measuring the K_s in situ, (2) compare the differences in infiltration capacity, preferential flow, and soil properties and structure between the rubber tree planting

zone (RT) and the tea tree planting zone (TT), and (3) identify the water supply mechanisms and implications for the sustainable and safe management of water resources security.

2. Materials and methods

2.1. Study site

The study site was located in the Xishuangbanna Tropical Botanical Garden (21°55'39"N, 101°15'55"E) in the Yunnan Province, SW China (Fig. 1). The experiments were carried out in a small typical catchment (19.2 ha) that consisted of rubber monoculture and rubber-tea agroforestry systems. The elevation ranged from 370 m to 680 m above sea level, with gentle slopes of 10–16°. The local climate is dominated by tropical southwest monsoon from the Indian Ocean from May to October (rainy season) and by subtropical jet streams from November to April (dry season). Climate records from a nearby weather station show that the mean annual precipitation is 1475 mm, with 85% occurring in the rainy season and 15% occurring in the dry season. The annual mean air temperature is 21.7 °C (Liu et al., 2016).

Rubber trees (*Hevea brasiliensis*) were widely intercropped with tea plants (*Camellia sinensis*) in this area. A 12.0 m × 14.5 m plot was established in the rubber-tea agroforestry system, including three rows of tea and two rows of rubber (Fig. 1). Rubber trees were planted with a 2.0 m × 4.5 m spacing on bench terraces after complete clear-cutting of the naive rainforest in the late 1980s. Each set of double rows was separated by a 9.0-m wide gap to allow for intercropping. Three rows of tea were planted in the middle of the gap at a density of 0.2 m × 3.0 m in 1997. The tea trees on the riser banks of the gap have not been manually pruned. After twenty-two years of growth, they were 2.5 m–5.0 m high.

Detailed information about the vegetation and soil in the study site are shown in Table 1. The leaf area index ($\text{m}^2 \text{m}^{-2}$) and canopy cover (%) were measured using an LAI-2200 plant canopy analyser (Li-Cor Inc., USA). Crop residues were randomly collected from six quadrants (1.0 m × 1.0 m) on the soil surface. Twelve litter samples were collected and oven-dried at 65 °C and weighed. For the belowground root biomass (0–30 cm), the roots were collected near each stainless steel cylinder used to determine the K_s using a soil drill (inner diameter 4.5 cm, height 25.0 cm). Sixty-six root biomass samples were collected. After washing off the soil in a 2 mm sieve, the root samples were oven-dried at 105 °C to a constant weight. The remaining soil samples were air-dried and the organic matter content and total nitrogen were measured using the CN-analyser (Vario MAX CN, Elementar Analysensysteme GmbH, Germany). Due to the low soil moisture and negligible faunal activity in the dry season (Fonte et al., 2010), earthworms were

randomly sampled at each zone of the study site in July 2018. Twelve soil cores (10 × 10 × 30 cm deep) were rapidly excavated and the earthworms were hand-sorted in the field. The earthworms were returned to the lab, where they were cleaned and weighed after wiping dry with paper.

2.2. Saturated hydraulic conductivity (K_s)

The measurements of field K_s were conducted within the study plot (12.0 m × 14.5 m, Fig. 1) from March 12 to April 10, 2018, during which no rainfall occurred in this area. Sixty-six stainless steel cylinders (inner diameter 20.0 cm, height 15.0 cm, wall thickness 0.2 cm) were carefully driven 5.0 cm into the soil, minimizing the disturbance of the soil porous system. The sidewalls and edges of cylinders were water tight. Each cylinder was initially filled with tap water equivalent to a water head of 10.0 cm, which was controlled by a reference ruler, and the reduced water level height was recorded at 2-min intervals for 30 min and at 5-min intervals after 30 min. The initial infiltration rate (IIR, cm min^{-1}) was defined as the ratio of the reduced water depth (cm) to the initial infiltration time (2 min). Each cylinder was rapidly refilled to a height of 10.0 cm after each recording until the difference in water level between filling and after 5 min remained the same for five consecutive times. The single-ring infiltration approach with large ponding depth can help to increase the accuracy of flow-rate measurements compared to the small ponding depth, and also decrease overall measurement time in low-permeability materials (Reynolds and Elrick, 1990). The refilling procedure took approximately 1.7 h. We assumed that a steady-state flow occurred at this point, and the actual steady-state infiltration rate (I_s , cm min^{-1}) was calculated based on the last five measured values (Bodhinayake et al., 2004). Based on the relationship between water infiltration and water temperature (Yang et al., 2017), the I_s was converted to the quasi-steady infiltrate rate (SIR, cm min^{-1}) at 10 °C according to Eq. (1), and the SIR was used to calculate the field hydraulic conductivity of saturated soil (K_s , cm min^{-1}) in accordance with Eq. (2) (Reynolds and Elrick, 1990; Youngs et al., 1993):

$$SIR = \frac{I_s}{0.7 + 0.03T} \quad (1)$$

$$K_s = \frac{SIR}{\frac{H_s}{C_1 d_s + C_2 r_s} + \frac{1}{S(C_1 d_s + C_2 r_s)} + 1} \quad (2)$$

where SIR and I_s are the infiltration rate under 10 °C and actual water temperature ($t = 20$ °C), respectively. $C_1 = 0.316\pi$ and $C_2 = 0.184\pi$ are dimensionless constants. H_s is the height (10.0 cm) of water inside the cylinder (i.e., the water height above the ground surface level), d_s is the insertion depth of the cylinder, r_s is the radius of the cylinder, and S

Table 1

General characteristics of vegetation and soil for the rubber planting and tea planting zones in the rubber-tea agroforestry system in Xishuangbanna.

Item		Rubber planting zone	Tea planting zone
Vegetation	Height of trees (m)	24.13 ± 1.78a	4.25 ± 0.59b
	Basal diameter (cm)	33.48 ± 1.87a	2.86 ± 0.37b
	Crown breadth (m)	10.18 ± 0.77a	3.02 ± 0.29b
	Leaf area index ($\text{m}^2 \text{m}^{-2}$)	1.33 ± 0.24b	3.13 ± 0.30a
	Canopy cover (%)	66.83 ± 4.37b	81.17 ± 1.66a
	Stand litter (kg m^{-2})	0.43 ± 0.03b	0.55 ± 0.08a
Soil (0–30 cm)	Clay (%)	33.7 ± 0.8a	23.1 ± 0.6b
	Silt (%)	34.4 ± 2.0a	35.3 ± 0.9a
	Sand (%)	31.9 ± 1.3b	41.6 ± 1.5a
	Percentages of gravel (%)	25.4 ± 1.1b	33.7 ± 1.2a
	Organic matter content (g kg^{-1})	17.2 ± 0.3b	19.7 ± 0.3a
	Total nitrogen (g kg^{-1})	2.04 ± 0.02b	2.30 ± 0.04a
	Root biomass (kg m^{-2})	0.56 ± 0.05b	0.71 ± 0.07a
	Earthworm density (g m^{-2})	5.31 ± 0.68a	7.36 ± 0.51a
	Number of ant holes (diameter > 1.0 cm)	9.83 ± 0.70a	12.50 ± 1.02a

Data are expressed as the mean ± standard error (n = 6). Different lowercase letters indicate a significant difference at $P < 0.05$. Soil fractions (sand, silt, clay) is based on *Keys to Soil Taxonomy* (USDA-SCS, 1994).

is sorptive number (Reynolds, 2008) and this number indicates the relation between gravity and capillary forces in soil (Reynolds and Elrick, 1990). For this study, H_s , d_s , r_s , and S were 10, 5, 10 cm, and 0.12 cm^{-1} , respectively. The H_s for each run was measured from the reference rule to the top of the organic soil horizon.

2.3. Dye tracer infiltration

Two dye tracer infiltration experiments were conducted in the rubber and tea tree zones in the study plot (Fig. 1) to visualize the infiltration pattern and water flow paths. At each test position, the freshly fallen leaves were removed from the surface, leaving behind an undisturbed litter layer with older leaves on the ground. Then, a 30.0-cm high single hollow stainless steel cylinder (inner diameter 20.0 cm) was embedded 5 cm into the soil and its sidewalls and edges were water tight. Twenty-five litres of Brilliant Blue FCF dye solution at a concentration of 4.0 g L^{-1} (Flury and Flühler, 1995) were gradually added to the cylinder in selected time intervals as K_s measurement procedure. The value of pressure head was 10 cm and total infiltration time was recorded when the solution disappeared. The cylinder quadrants were then covered with large plastic sheets to avoid solution evaporation and dilution by rainfall. The plastic sheets and cylinders were removed 24 h after the end of infiltration. An 80 cm (width) $\times$ 120 cm (depth) dye-stained vertical soil profile was carefully excavated at the centre of the cylinder using a spatula. Tree roots, cracks, passages and holes produced by earthworms and ants were observed and counted during the excavation. Calibrated steel tapes were placed on the pits' soil walls before photographing to facilitate image analysis. Two vertical soil profiles were photographed using a Canon digital camera (Canon EOS M6, Japan) under a soft light condition provided by a black umbrella. The distance between camera and the centre of the vertical soil profile was approximately 50 cm. Dye-stained calibration patches were prepared for the imaging of soil horizons to define concentration categories for dye-stained soil regions during image analysis (Weiler and Flühler, 2004).

According to the image processing procedures described by Cey and Rudolph (2009), the pictures from the soil profiles were geometrically corrected by using ERDAS IMAGINE 9.2 (Erdas Inc., Atlanta, Georgia, USA) and classified into dyed and non-dyed areas. Based on the dye staining intensity, the dyed areas were divided into three relative classes using a supervised classification (Bargués Tobella et al., 2014). The dye-stained soil calibration patches were used to estimate the concentration categories of the three dye-stained soil classes, 0.05–0.5, 0.5–2.0, and $> 2.0 \text{ g L}^{-1}$, respectively (Jiang et al., 2015). The concentration of Brilliant Blue FCF can imply the soil water content. Thus, we defined the water active zones as the dye-stained zones with a high dye-stained concentration of $> 2.0 \text{ g L}^{-1}$ where preferential flow dominated. The dye-stained zones with low dye-stained concentration ($0.05–2.0 \text{ g L}^{-1}$) were defined as the water buffer zones where the water exchange between the preferential flow and the soil matrix occurred. The area (cm^2) of each dye-stained concentration category was calculated using the *Tabulated area tool* in ArcMap 10.3 (ESRI Inc., Redlands, California, USA). This tool calculated the area covered by each class within different zones in the dye-stained image, which were defined by a polygon shapefile (Bargués Tobella et al., 2014). The polygon shapefile was divided into a grid with 60 rectangular 160 cm^2 polygons (80 cm wide $\times$ 2 cm high). A tapeline was used to measure the maximum dye-stained width and depth of the soil profile.

In addition, a 20 cm (width) $\times$ 120 cm (high) dye-stained image was separated from the total dye-stained image to calculate preferential flow indices. Image processing was conducted using ArcMap 10.3 in accordance with the procedures described above. Preferential flow indices were calculated from the output data and included the following parameters:

(a) Dye coverage (DC , %), defined as the percentage ratio of the dye-stained area to the sum of dye-stained and non-stained areas (Flury

et al., 1994).

$$DC = \left(\frac{D}{D + ND} \right) \times 100 \quad (3)$$

where DC is the dye coverage (%), D is the dye-stained area (cm^2), and ND is the non-stained area (cm^2).

(b) Uniform infiltration depth (UID , cm), calculated as the depth at which the DC decreases below 80%, indicating the depth to which matrix flow is prevalent. In general, soils with a low UID have a high degree of preferential flow (Benegas et al., 2014; Van Schaik, 2009).

(c) Preferential flow fraction ($PF\text{-}fr$, %), defined as the fraction of the preferential flow area to the sum of the dye-stained areas as follows (van Schaik, 2009):

$$PF\text{-}fr = \left(1 - \frac{UID \times W}{SA} \right) \times 100 \quad (4)$$

where $PF\text{-}fr$ is the preferential flow fraction (%), UID is the uniform infiltration depth (cm), W is the soil profile width (20 cm in this study), and SA indicates the stained area (cm^2).

(d) Length index (LI), defined as the sum of the absolute differences between subsequent DC values with depth in a profile (Bargués Tobella et al., 2014; Mei et al., 2018). Soils with a high LI have a high degree of preferential flow. It is calculated as:

$$LI = \sum_{i=1}^{60} |DC_{i+1} - DC_i| \quad (5)$$

where LI is the length index, and i is the depth interval of the rectangular area in which dye coverage (DC , %) was calculated.

2.4. Soil physical characteristics

Soil physical properties were determined using bulk soil samples from the study site where no rainfall occurred for half a month (Chen, 2005; Jiao et al., 2011). In the dye tracer experiment sites, bulk soil samples (three replicates) were randomly collected at depths of 5, 15, 25, 35, 45, 55, 65, 75, 85, 95, 105, and 115 cm with pre-weighed (weight: W_{PRCC} , g) cutting cylinders (inner diameter 7.0 cm, height 5.2 cm, volume 200 cm^3). The mean value of a certain soil property at each depth was used to represent the overall attributes of the study plot. In the laboratory, the cutting cylinders containing wet soil (W_{CCWS} , g) were saturated from below by placing them in distilled water. The water was nearly at the top edge of the cutting cylinders, ensuring no water entered the soil samples from above. The weight of the soil samples at saturation was determined (W_{SAT} , g) after the soil core cylinders were ponded for 24 h. Then, the saturated soil core cylinders were placed on a dry sand layer, and the weight of water drained through gravity was measured after 2 h and 5 days (W_{2H} and W_{5D} , g), respectively. The weight of the cutting cylinders containing dry soil was determined (W_{CCDS} , g) after oven drying at 105°C for 48 h. The total porosity was calculated in undisturbed water-saturated 200 cm^3 samples with the assumption that no air was trapped in the pores and validated using the dry bulk density and a soil particle density of 2.65 g cm^{-3} (Danielson and Sutherland, 1986). The soil physical properties were calculated using the following formulae:

$$\text{Initial gravimetric water content} = \frac{W_{CCWS} - W_{CCDS}}{W_{CCDS} - W_{PRCC}} \times 100\% \quad (6)$$

$$\text{Bulk density} = \frac{W_{CCDS} - W_{PRCC}}{200} \quad (7)$$

$$\text{Total porosity} = \left(1 - \frac{\text{Bulk density}}{\text{Soil particle density}} \right) \times 100\% \quad (8)$$

$$\text{Capillary holding capacity} = \frac{W_{2H} - W_{CCDS}}{W_{CCDS} - W_{PRCC}} \times 100\% \quad (9)$$

$$\text{Capillary porosity} = \frac{\text{Bulk density} \times \text{Capillary holding capacity}}{\rho_{\text{water}}} \quad (10)$$

$$\text{Noncapillary porosity} = \text{Total porosity} - \text{Capillary porosity} \quad (11)$$

$$\text{Saturated water capacity} = \frac{W_{\text{SAT}} - W_{\text{CCDS}}}{W_{\text{CCDS}} - W_{\text{PRCC}}} \times 100\% \quad (12)$$

$$\text{Field water capacity} = \frac{W_{\text{SD}} - W_{\text{CCDS}}}{W_{\text{CCDS}} - W_{\text{PRCC}}} \times 100\% \quad (13)$$

$$x_s = 100\% - \text{Total porosity} \quad (14)$$

$$x_l = \frac{\text{Initial gravimetric water content} \times \text{Bulk density}}{\rho_{\text{water}}} \quad (15)$$

$$x_g = \text{Total porosity} - x_l \quad (16)$$

$$\text{TPSI} = [(x_s - 25)x_l x_g]^{0.4769} \quad (17)$$

Where TPSI represents the three soil phase index (TPSI, %) ranging from 0 to 100, where x_s , x_l , and x_g is the volumetric solid content (%), volumetric water content (%), and volumetric gas content (%). A higher TPSI indicates a better soil structure (Wang et al., 2015).

2.5. Statistical analysis

All data were checked for normal distribution and homogeneity of variances via the Kolmogorov-Smirnov test and Levene's test, respectively. A log transformation or square-root transformation was conducted when the assumptions were not met. One-way ANOVA followed by a Tukey test for multiple comparisons between means were applied to assess the differences in IIR , SIR , and K_s between the rubber tree, transition, and tea tree zones. An independent-Samples t -test was used to evaluate the differences in soil and vegetation characteristics between the rubber and tea planting zones. All statistical procedures were performed with an $\alpha = 0.05$ threshold for significance in IBM SPSS version 19.0 statistical software (SPSS Inc., Chicago). Spearman's correlation coefficients were applied to determine the correlation among the soil physical properties. Ordinary kriging was used to generate distribution maps for K_s at different positions. The contour map of K_s was drawn using Golden Software Surfer 12 (Golden software Inc.).

3. Results

3.1. Spatial variation of K_s and soil water infiltration rates

The isogram of the K_s presents the spatial distribution of the soil hydraulic conductivity in different zones of the rubber-tea agroforestry system (Fig. 2(a)). The K_s in the TT was highest (on average: 0.37 cm min^{-1}) and ranged from 0.15 to 0.75 cm min^{-1} and the lowest K_s (on average: 0.14 cm min^{-1}) was observed in the RT (range: 0.0011 – 0.50 cm min^{-1}). The average K_s in the transition zone (between rubber trees and tea trees, TZ: 0.23 cm min^{-1}) was between the K_s of the TT and RT. The K_s values of the soil between two rubber trees in a row were lower than those between rubber tree rows (Fig. 2(b)) whereas the K_s of the soil close to tea trees was higher than that between tea rows. In summary, the K_s decreased with increasing distance from the nearest tea trunk. The K_s values between tea rows were also higher than those between rubber rows. However, K_s exhibited a high spatial variability. Individual K_s measured in the TT was not always higher than that observed in the RT.

The soil water infiltration process took approximately 105 min in this study. The infiltration rates for all three zones rapidly decreased within the first 10 min and did not change after 60 min (Fig. 3(a)). The IIR of the soil was 2.84 cm min^{-1} and was significantly higher than that in the TZ (1.47 cm min^{-1}) and the RT (0.95 cm min^{-1}). Similarly, the SIR of the soil in different zones of the rubber-tea agroforestry system were:

TT (0.65 cm min^{-1}) > TZ (0.40 cm min^{-1}) > RT (0.25 cm min^{-1}).

3.2. Water flow behaviour in soil

The types of water flow were interpreted directly from the two vertically dye-stained patterns cut along the centre line of the stainless steel cylinder (Fig. 4(a) and (c)). Preferential flow, which includes macropore flow in all directions and lateral flow, was the dominant flow type for the two experimental zones. The dye was distributed uniformly throughout the 0–40 cm soil depth whereas the dye-stained pattern exhibited randomly dispersed patches below 40 cm. The distribution of the dye mainly occurred in the 0–90 cm soil layer in the RT and 0–120 cm in the TT. The stain characteristics in Table 2 show that the total dye-stained area in RT (2720.5 cm^2) was lower than that in TT (3227.4 cm^2). The RT soils took more time (60 min) to infiltrate 25.0 L water than the TT (35 min). The maximum width and depth of dye-stained patterns in the RT were 82.3 cm and 112.6 cm, respectively; they were slightly lower than the 87.5 cm and 128.1 cm in the TT. In the RT and TT, the proportion of the water active zones to the total stained area was 19% and 28%, respectively; the proportion of the water buffer zone to the total stained area was 81% and 72%, respectively. Furthermore, the proportion of the water active zone was clearly lower than that of the water buffer zone.

The quantitative infiltration behaviour information illustrates the variation in the classified dye-stained area in relation to the soil depth (Fig. 4(d) and (g)). In both the RT and TT zones, the dark blue area with high dye concentrations ($> 2.0 \text{ g L}^{-1}$) initially decreased slightly, increased, decreased, increased again and finally decreased from the surface to 60 cm. Approximately 94% and 86% of the water active zone were (i.e., the dye-stained zones with a concentration of $> 2.0 \text{ g L}^{-1}$) confined to the 0–45 cm of the soil profile in the RT and 0–35 cm in the TT, respectively. The water buffer zone (i.e., the dye-stained zones with a concentration of 0.05 – 2.0 g L^{-1}) in the RT generally distributed throughout the entire soil profile. For the TT, a larger proportion of water buffer zones with the concentration of 0.5 – 2.0 g L^{-1} (77%) and 0.05 – 0.5 g L^{-1} (61%) was confined to the soil layers of 0–50 cm and 40–80 cm, respectively.

All six parameters used in this study to quantify the degree of preferential flow showed consistent trends: the preferential flow was higher in the TT than in the RT (Fig. 5). The ID , DC and SA from the $20 \text{ cm} \times 120 \text{ cm}$ dye-stained patterns in the RT (99.1 cm , 50.2% and 1205.9 cm^2 , respectively) were lower than those in the TT (they were 120.0 cm , 62.5% and 1501.1 cm^2 , respectively). The UID , below which the preferential flow is expected to initiate, was 43.0 cm in the RT, higher than that in the TT (37.2 cm). Both the measured LI and PF were lower in the RT than in the TT.

3.3. Soil physical properties and structure

Soil physical characteristics and structure for two sample zones of the rubber-tea agroforestry system are shown in Fig. 6. The bulk density was consistently higher (except at the 105–115 cm depth) in the RT than in the TT. Averaged over a depth range of 0–120 cm, the noncapillary porosity, gas phase, and TPSI in the RT were 1.4%, 6.3% and 67.7%, respectively, and were significantly lower than those in the TT, which were 3.1%, 9.6%, and 78.5%, respectively. The noncapillary porosity in the TT initially increased and then decreased with the soil depth, and then increased and decreased again. The capillary porosity in the TT was lower than that in the RT below the 10 cm soil depth. Soil organic matter, total nitrogen, root biomass in the TT were significantly higher than those in the RT (Table 1). Higher earthworm density and larger number of ant holes were observed in the soils beneath tea trees than those below rubber trees. All the soil physical properties were correlated at different levels (Table 3).

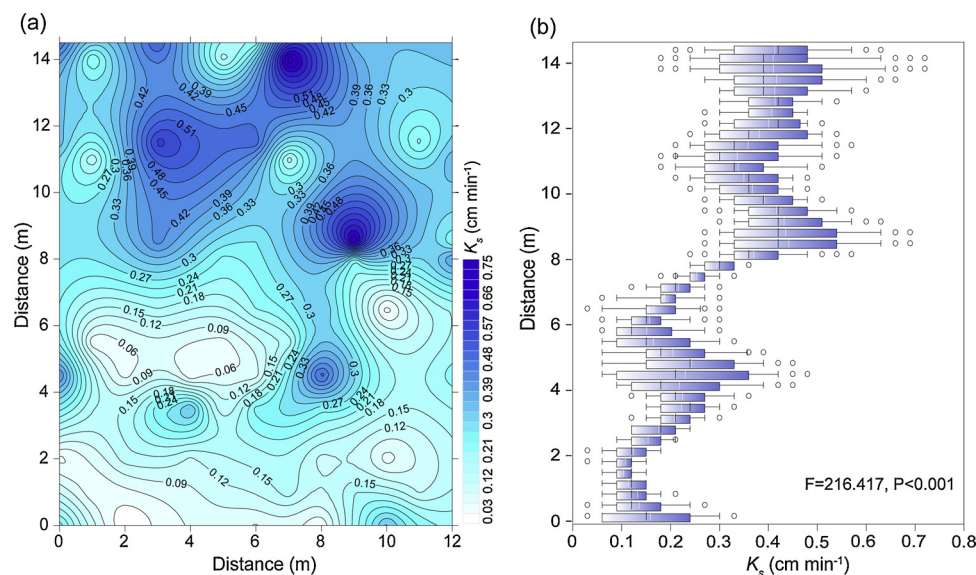


Fig. 2. Spatial variation (a) of saturated hydraulic conductivity (K_s , cm min^{-1}) and boxplots of K_s values (b) in the rubber-tea agroforestry system in Xishuangbanna.

4. Discussion

4.1. Spatial variation of saturated soil hydraulic conductivity

The K_s expresses the ability of a soil to conduct water under saturated conditions (Jačka et al., 2014), which determines whether rainfall infiltrates into a soil or flows along the surface (Zwartendijk et al., 2017) and consequently affects water flow and transport of dissolved solutes (Jačka et al., 2014). The K_s generally varies with different field conditions, soil structure, and physical properties (Reynolds et al., 2000). In our study site, the K_s was measured on undisturbed soil in a rubber-tea agroforestry system. The hydraulic conductivity values exhibited a high spatial heterogeneity (Fig. 2). The K_s decreased with the distance from the trunks of tea trees and the K_s values between tea trees within a row were higher than that between tea rows. These results suggest that the tea trees have a clear positive impact on the soil infiltrability in the rubber-tea agroforestry system. This is consistent with previous studies. For example, Chandler and Chappell (2008) noted that K_s in a parkland in northwest England decreased away from oak (*Quercus robur*) trees. In contrast, the K_s measurements between rubber trees in a row were generally lower than those between the rubber rows. Undoubtedly, the differences in K_s values resulted in an increase in spatial heterogeneity of the soil water infiltrability in rubber-based agroforestry systems. Similarly, Benegas et al. (2014) found that trees in a coffee agroforestry system in Central America had no significant effect

on soil infiltrability. It is possible that the soil on terrace benches in our study was severely compacted by daily land management such as trampling, latex tapping and herbicide application. The denser structure of the topsoil layer caused by land use is characterized by higher soil bulk density and lower total porosity (Fig. 6; Liu et al., 2016; Zhu et al., 2018), which exert a very localized negative effect on K_s (Blevins et al., 1970; Chandler and Chappell, 2008; Whalley et al., 2004). Reduced K_s can limit water flux into the underlying soil matrix and consequently counteract the positive effect of rubber trees on the K_s .

In addition, the soils in the tea planting zone were less degraded and had higher field infiltration abilities (e.g., IIR and SIR) than those in the rubber planting zone (Fig. 3). If the rubber planting zone was regarded as rubber monoculture, our findings agree with previous research in Costa Rica showing that the level of water infiltration in a coffee agroforestry was greater than that in a coffee monoculture (Cannavo et al., 2011). This result may be explained by the improved soil structure and soil properties beneath tea trees, which are closely related to the higher soil organic content and increased soil biotic activity and root activity (Table 1; Bargués Tobella et al., 2014; Berglund et al., 1981; Lehmann et al., 2001; Zhu et al., 2018). The tea trees at this site had established perennial vegetation for 22 years, with extensive root systems and higher root biomass. Root decay and larger quantities of leaf residues increased soil organic input beneath the trees. Higher earthworm density and considerable ant holes in the tea planting area, coupled with more soil organic matter (Table 1), contributed to the

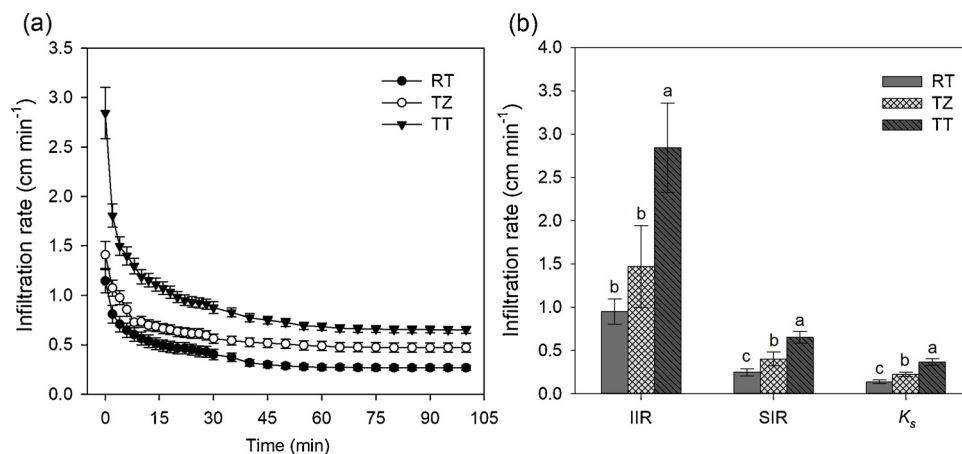


Fig. 3. (a) Variation in the infiltration rate with time, and (b) initial infiltration rate (IIR, cm min^{-1}), quasi-steady infiltration rate (SIR, cm min^{-1}), and saturated hydraulic conductivity (K_s , cm min^{-1}) for three different zones of the rubber-tea agroforestry system in Xishuangbanna. RT: Rubber tree planting zone, TZ: Transition zone, TT: Tea tree planting zone. Data are expressed as the mean $\pm$ standard error ($n = 8-37$). Different lowercase letters indicate a significant difference at $P < 0.05$.

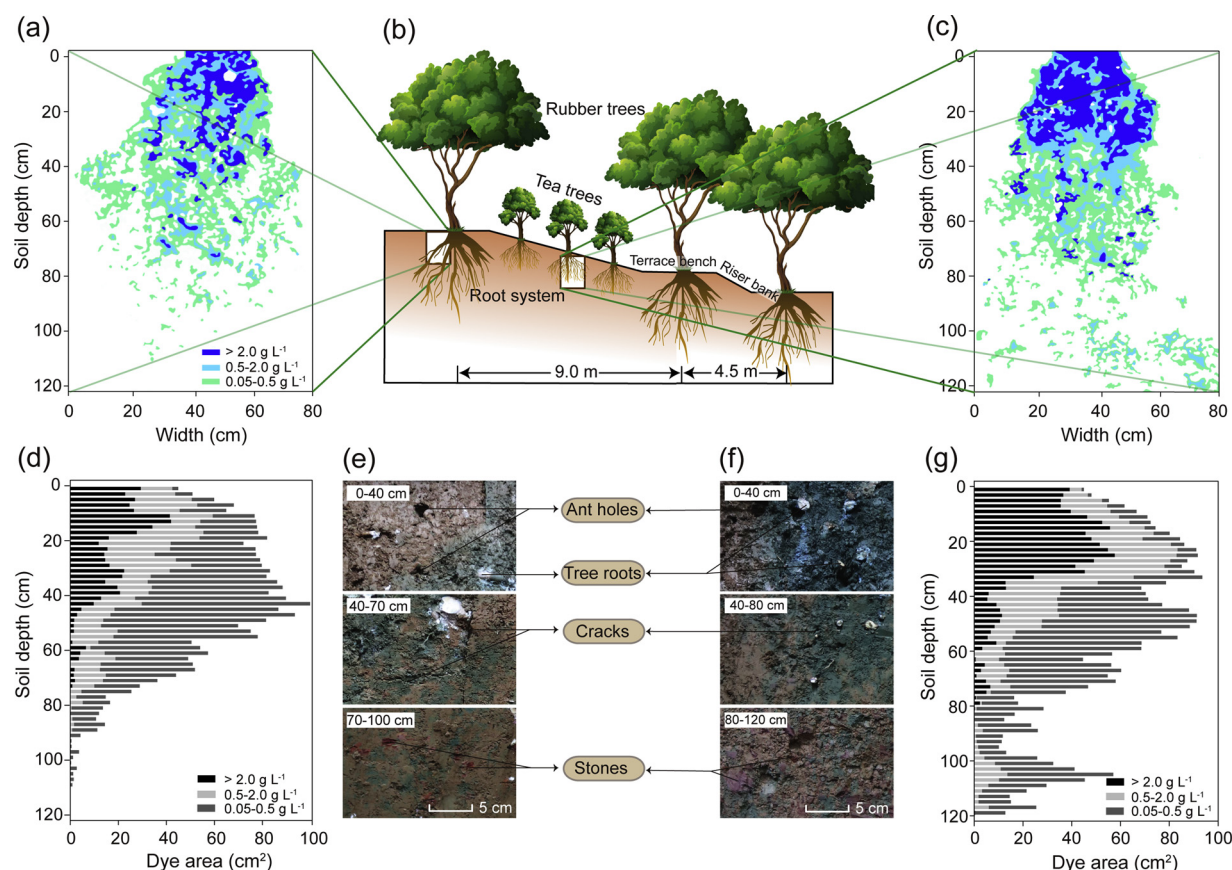


Fig. 4. Infiltration patterns for vertical soil sections in the rubber tree planting zone (RT) (a) and tea tree planting zone (TT) (c) under a rubber-tea agroforestry system (b). Calculated area (cm²) of the stained soil profile vs. depth (cm) for RT (d) and TT (g). Real structure of different soil layers for RT (e) and TT (f).

Table 2

Dye stain characteristics in the vertical dye soil profile for the rubber and tea planting zones in the rubber-tea agroforestry system.

Position	Infiltration time (min)	Maximum dye-stained width (cm)	Maximum dye-stained depth (cm)	Total stained area (cm ²)	Area (cm ²) of concentration		
					> 2.0 g L ⁻¹	0.5–2.0 g L ⁻¹	0.05–0.5 g L ⁻¹
Rubber planting zone	60.0	82.3	112.6	2720.5	519.6	666.8	1534.1
Tea planting zone	35.5	87.5	128.1	3227.4	899.9	789.7	1537.8

formation of soil macropores and positively impacted aggregate stability, improving soil structure and soil hydraulic conductivity (Bargués Tobella et al., 2014; Chandler and Chappell, 2008; Mordelet et al., 1993). Extreme soil desiccation during the dry season can also open structural cracks in soils (e.g., Fig. 3 (e)–(f)) and increase the infiltration capacity of occasional precipitation. Moreover, tea shrubs with a low and dense canopy can substantially protect the soil surface from rain-drop impact, increasing the soil stability and reducing the development of soil crusts that largely limit water infiltration (Young, 1989; Liu et al., 2016). By filtering splashed soil particles, a thicker litter layer beneath tea trees could prevent the clogging of soil pores, resulting in an increased K_s (Zhu et al., 2018).

4.2. Preferential flow pathways and their impacting factors

Dye tracers provide visual information on water infiltration along preferential flow pathways in the soil profile (Flury and Wai, 2003). Calculated preferential flow parameters (e.g., ID, DC, UID, SA, PF-fr, and LD) based on dye images can assess the extent of preferential flow. Along with the preferential flow parameters (Fig. 5), the infiltration time and dye stain characteristics (Table 2) demonstrate that the degree of preferential flow was obviously higher in the TT than that in the RT,

with a larger fraction of the infiltrating water channelized through preferential pathways. In addition, a higher UID in the RT indicated that the water flowed uniformly through the soil matrix down to deeper depths in contrast to the TT, where the preferential flow was initiated earlier as shown by the shallower UID. Preferential water flow behaviours are generally influenced by the interactions among root systems, faunal activities, and soil properties (Bargués Tobella et al., 2014; Benegas et al., 2014; Beven and Germann, 2013). In our study, the rubber trees had deeper and wider root systems whereas the tea trees had lateral roots less than 1 m (Beer, 1987; Niranjana and Viswanath, 2008). Abundant tea roots in the surface soil and rubber roots in the deep soil (extending more than 1.2 m deep) under the intercropping area produced more preferential flow paths by root decay and penetrating the soil matrix (Devitt and Smith, 2002). More earthworm and termite activity in soils beneath tea trees can generate more macropores (e.g., tunnels and burrows) and improve macropore flow, which is a subset of preferential flow (Alaoui et al., 2011). Furthermore, a lower soil bulk density, higher field capacity and soil porosity, and inhomogeneous cracks, fissures, and small stones in soils also lead to higher preferential flow in the TT (Bromley et al., 1997).

Because of the hard and firm soil, the infiltration time in the RT was almost twice that in the TT (Table 2). This result suggested that the

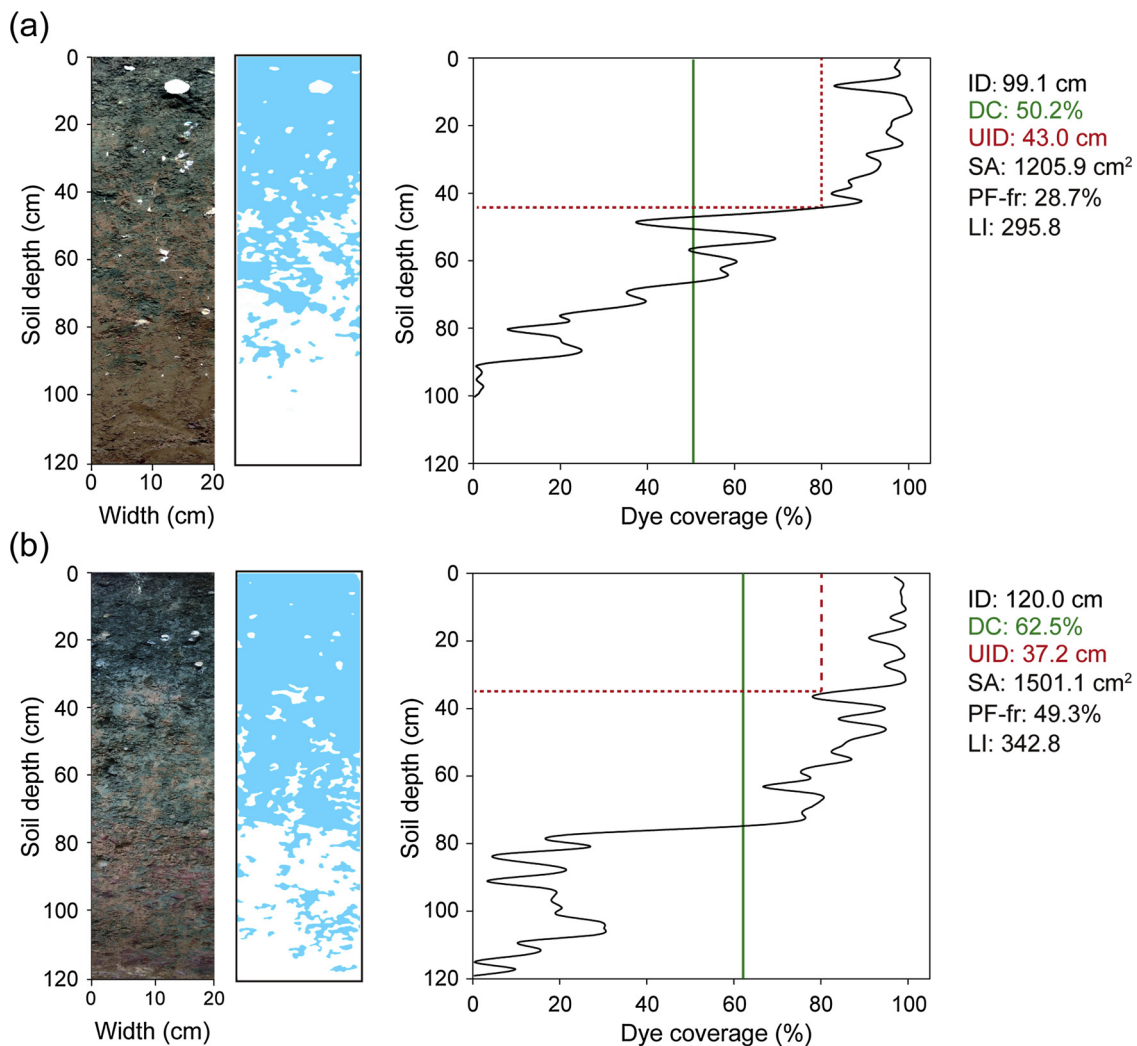


Fig. 5. The 20-cm-wide stained profiles for the rubber planting zone (a) and tea planting zone (b) and their corresponding dye coverage profiles. The parameters used to measure the degree of preferential flow are shown next to each picture. ID: Infiltration depth (cm), DC: Dye coverage (%), UID: Uniform infiltration depth (cm), SA: Stained area (cm²), PF-fr: Preferential flow fraction (%), LI: Length index. The intersection of the dotted lines and dye coverage (black lines) is the uniform infiltration depth.

loose soil with high macroporosity in the TT allows for rainwater to infiltrate faster into the soil during rainfall. The maximum dye-stained depth and width in the TT indicated that root channels, faunal tunnels, and cracks contributed to water infiltration into the deeper soil layer and extended water distribution in horizontal soil. Further, the dye-stained width across the total soil section was generally greater than the diameter of the stainless steel (20.0 cm), demonstrating that a large number of lateral roots can promote lateral flow and water exchange in the soil matrix. The improved preferential flow, lateral flow and water exchange resulted in the vertical and lateral redistribution of soil moisture, which can diversify the water supply; this feature is beneficial for the plant growth during a long dry season (November–April) in Xishuangbanna.

Both the water active zone (including the preferential flow paths) and water buffer zone were higher in the TT than in the RT; and the water buffer zone was always larger than the water active zone. The water active zone in the RT was mainly distributed in the upper 45.0 cm of the soil profile, which was heavily relied on by rubber trees during the rainy season (Wu et al., 2016). The water active zone in the TT was limited to the surface soil (< 30.0 cm). This surface soil water was mainly absorbed by tea trees with shallow roots (Niranjana and Viswanath, 2008; Wu et al., 2016). Moreover, a large proportion of the water buffer zone was confined to the 40–100 cm soil profile in both RT

and TT. The water buffer zone in the TT extended to 120 cm soil layer (Fig. 4(c)). This deep soil water could be utilized by rubber trees with long root systems during the dry season. Therefore, the storage and redistribution of soil water is enhanced by the intercropped tea shrubs. Different water sources of different soil layers in a rubber agroforestry system can reduce the interspecific competition for water. Considering the root characteristics, we suggest that land management should combine deep-root rubber trees with shallow-root crops; crops with more horizontal roots than vertical would compete less for resources with rubber trees.

4.3. Implications for watershed management

In Xishuangbanna, large areas of rubber plantations and intensive agricultural activities have resulted in low streamflow and water scarcity during the dry season and excess surface runoff and soil erosion during the rainy season at the watershed scale (Liu et al., 2016; Qiu, 2009; Tan et al., 2011; Zhu et al., 2018). In general, the marked differences in soil infiltrability and preferential flow among different zones in rubber-based agroforestry systems can lead to the spatial redistribution of surface and belowground water. The non-intercropping area (e.g., rubber planting zone) with low infiltrability and preferential flow acts as a source of surface runoff whereas the intercropping area

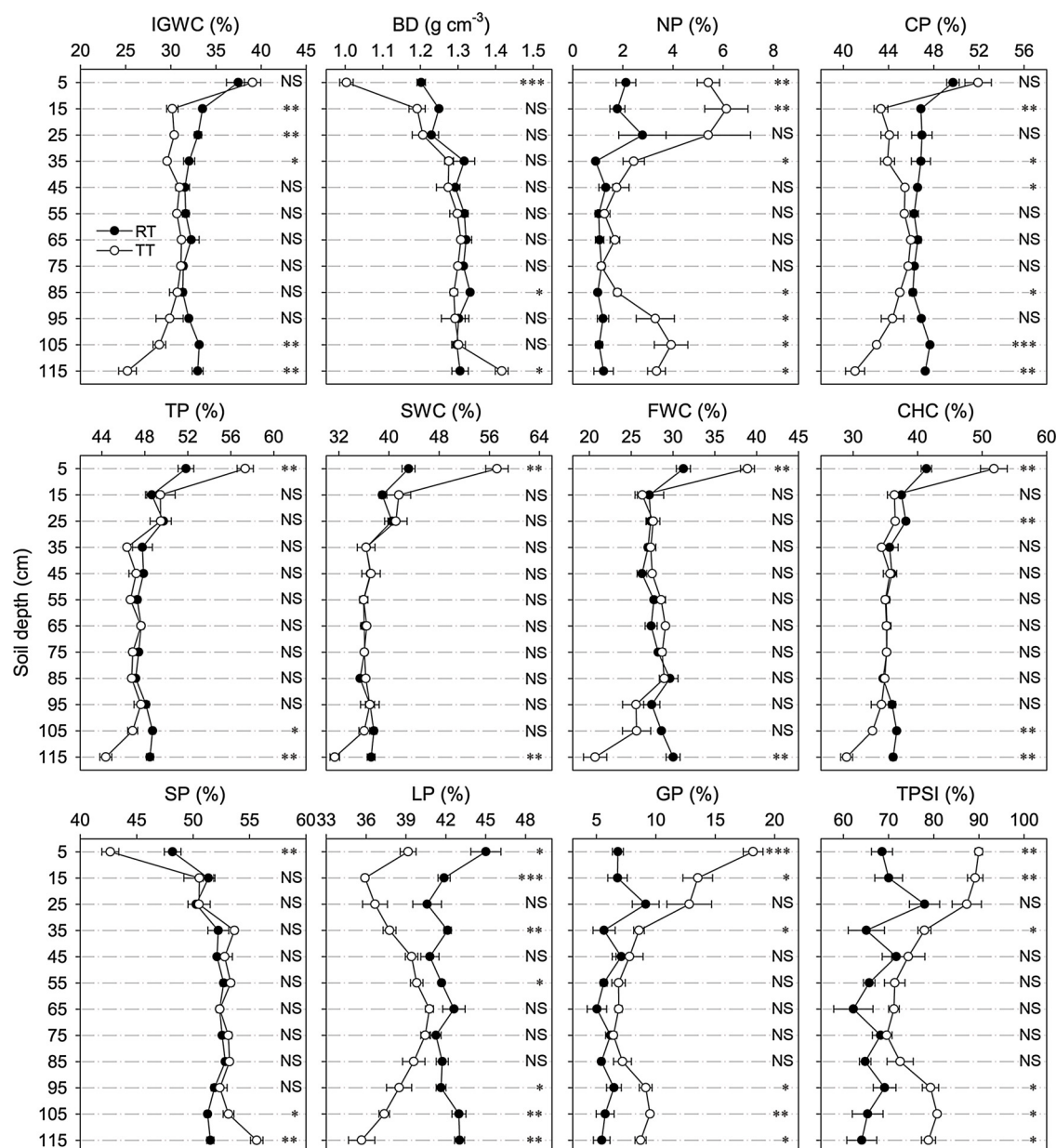


Fig. 6. Differences in physical properties at each soil layer between the rubber tree planting zone (RT) and tea tree planting zone (TT) in the study site. *IGWC*: Initial gravimetric water content, *BD*: Bulk density (g cm^{-3}), *NP*: Noncapillary porosity (%), *CP*: Capillary porosity (%), *TP*: Total porosity (%), *SWC*: Saturated water capacity (%), *FWC*: Field water capacity (%), *CHC*: Capillary holding capacity (%), *SP*: Solid phase (%), *LP*: Liquid phase (%), *GP*: Gas phase (%), *TPSI*: Three soil phase index (%). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, NS Not significant.

(e.g., tea planting area) with high infiltrability and preferential flow acts as a sink. Being a source implies the possibility of having lower infiltration rates, whereas to be a sink refers to higher infiltration rates. Thus, the improvement of infiltrability and preferential flow has a localized but beneficial effect on attenuating runoff generation by providing local infiltration zones beneath intercrops. Surface runoff from other areas also partially re-infiltrates into the soil when it flows across the intercropping area (Fig. 7), thereby increasing groundwater recharge and water availability for trees and intercrops (Liu et al., 2005; Bargaes Tobella et al., 2014). Furthermore, tea planting areas with more crop residues provided roughness elements that reduced the velocity of water flows and promoted infiltration (Chandler and Chappell, 2008). These could help explain why a rubber-tea agroforestry system in Xishuangbanna reduced surface runoff and sediment yield by 44% and 63%, respectively, compared with a rubber monoculture (Zhu et al., 2018). Reducing runoff and soil erosion losses concurrently limits

the transfer of agrochemicals (e.g., nutrients and pesticides) to surface water bodies, which potentially avoids decline in soil fertility and water quality (Li et al., 2018; Pavlidis and Tsihrintzis, 2018; Sidle et al., 2017). The shallower roots of tea trees can directly absorb nutrients that leach to the root zones and decrease the related water contamination. In addition, more organic matter inputs from litterfall and root decomposition, coupled with the sink of surface water, contribute to the development of the so-called “island of fertility” in rubber-based agroforestry systems. This supposition is supported by the fact that significantly higher soil organic matter and total nitrogen were observed in the tea planting zone than in the rubber planting zone (Table 1). Nutrient-rich processes in the intercrop planting zone are important for the growth of intercrops and rubber trees.

Diverse water supply mechanisms, multi-layered canopies, improved soil properties, and protective ground cover are favourable to the utilization and management of water resources in rubber-based

Table 3
Correlation coefficients of soil physical properties.

Indicator	IGWC	BD	NP	CP	TP	SWC	FWC	CHC	SP	LP	GP	TPSI
Initial gravimetric water content (IGWC)	1											
Bulk density (BD)		1										
Noncapillary porosity (NP)			1									
Capillary porosity (CP)				1								
Total porosity (TP)					1							
Saturated water capacity (SWC)						1						
Field water capacity (FWC)							1					
Capillary holding capacity (CHC)								1				
Solid phase (SP)									1			
Liquid phase (LP)										1		
Gas phase (GP)											1	
Three soil phase index (TPSI)												1

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

agroforestry systems (Fig. 8). However, there are potentially negative effects in such agroforestry systems. For example, nutrients, pesticides, and pathogens that cannot be entirely degraded by soil organisms or taken up by plants may rapidly bypass the soil matrix and transfer to deep soil layers with preferential flow, leading to the contamination of groundwater (Alaoui et al., 2011; Bowmer, 1987; Van Der Heijden et al., 2013). A reasonable seasonal distribution of water resources is important for the plant growth and survival in seasonally dry tropics. The water stocked in soil during the rainy season is utilized by rubber trees and intercrops and should be used and regulated effectively during the dry season. Therefore, more research on the long-term observation and quantification of distribution characteristics of water resources (both surface and underground) is needed in rubber-based agroforestry systems.

4.4. Limitation of the research

The single-ring approach is a convenient approach used for infiltration measurement; however, the selection of pressure head could be improved in future researches. High ponding depth used for infiltration tends to increase infiltration rate and therefore, among other parameters, values of K_s (Kutilek and Nielsen, 1994; Reynolds and Elrick, 1990). In other words, high values of pressure head should be avoided due to the possible influence of the porous system in the sampled soil, possible dominant flow in macropores, and overestimation of preferential flow (and overall value of K_s which is sum of K_s of macropores and K_s of soil matrix). This increasing effect can be an order of magnitude in soil with macropores due to possible pipe flow effect (like in a sink). Consequently, differences in infiltration characteristics (K_s , SIR, IIR) and preferential flow among TT and RT zones may be also overestimated due to different amounts of macropores. This overestimation should be avoided by using a pressure head of about 5 cm.

Although dye tracer experiments are commonly used to study water flow patterns, the dyed soil patterns do not match the water flow patterns exactly, because dissolved brilliant blues exhibits problematic sorption effects and lower mobility than untreated water in smaller-sized pores (Fér et al., 2016; Nobles et al., 2010; Wang and Zhang, 2011). In addition, the brilliant blue tests can advantageously reveal the fast component of the flow in macropores and upper size capillary pores, where the preferential flow occurs (Jačka et al., 2018; Schwen et al., 2014). Different dye tracers (i.e., potassium iodide and methylthionine chloride) may be used to study the water flow patterns in future researches.

Higher amounts of sand and stone were observed in the TT zone than in the RT zone, indicating that there is probably also a factor affecting infiltration and preferential flow other than only the cultivated crop. This factor can be moderately different initial soil conditions, concretely different content of gravel and different particle size distributions.

5. Conclusion

Rubber-based agroforestry continues to be a promising land-use practice in Xishuangbanna. Understanding its impacts on important water- and soil-associated hydrological ecosystem functions and services is imperative. The findings from this study demonstrate that intercropping tea trees with rubber trees positively impacts soil hydraulic properties, with higher infiltrability and preferential flow being in soils underneath tea trees. Long-term tea growth (22 years) significantly improved soil physico-chemical properties such as a lower bulk density, higher porosity, field water capacity, $TPSI$, and organic matter content. These sensitive improvements of soil properties, coupled with increased root and faunal activities, and other inhomogeneous structures (e.g., cracks and stones) in the tea planting zone substantially enhanced the infiltration capacity and preferential flow. The measured hydraulic

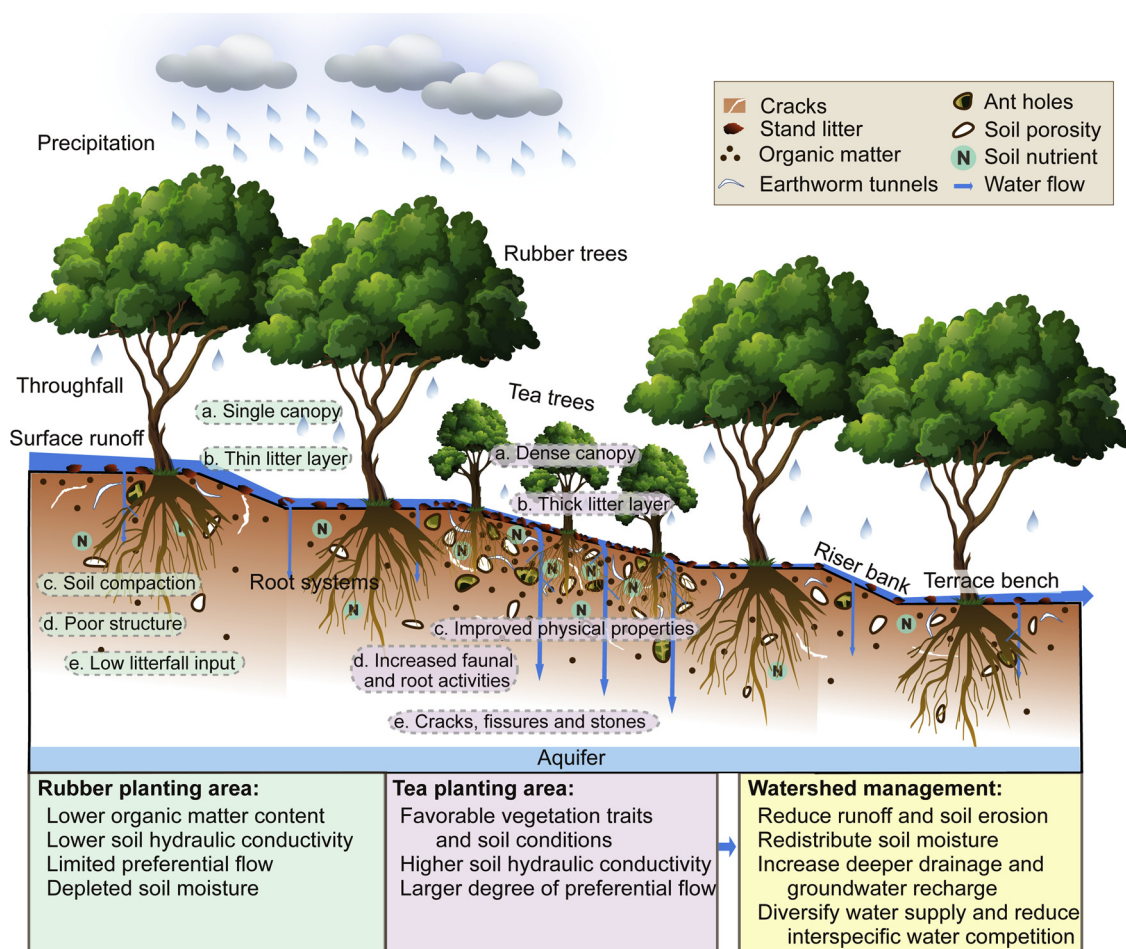


Fig. 7. Schematic diagram illustrating the rubber-tea agroforestry system for improving basin water resources management.

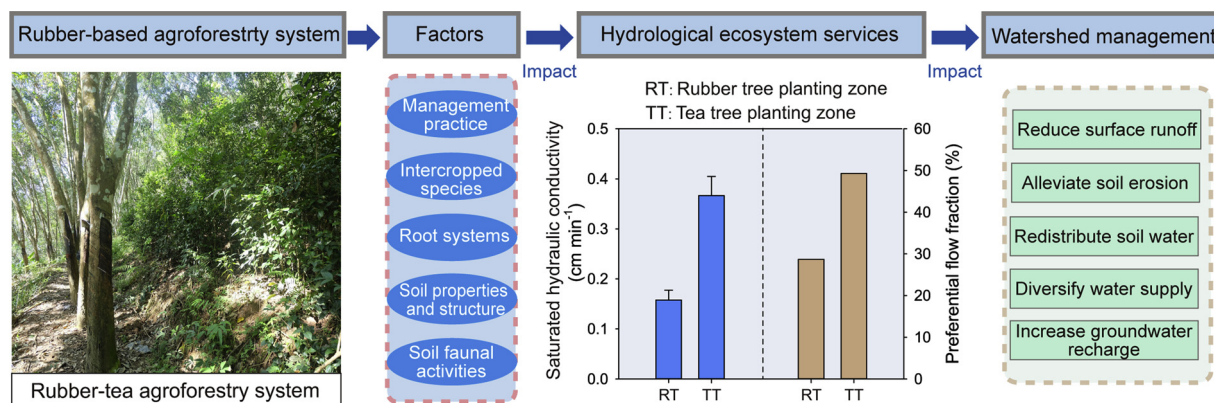


Fig. 8. Effects of intercrops on hydrological ecosystem services and watershed management in rubber-based agroforestry system.

conductivity showed significant spatial heterogeneity and decreased with the distance from the trunks of tea trees. The K_s between rubber trees on the terrace benches were lower than those on the riser banks between rubber rows. Intensive agricultural activities may have compacted the soil on terrace benches, thereby degrading the soil such as the losses in soil organic matter (Li et al., 2012), soil hardening and crusting (Liu et al., 2015), and excessive water and soil erosion (Liu et al., 2016; Zhu et al., 2018). The soil degradation in agriculture may reduce its infiltrability. Therefore, the rubber trees appeared to have less impact on the hydraulic conductivity. Local improvement of soil infiltrability and preferential flow in the rubber-tea agroforestry system could attenuate runoff generation and soil erosion, redistribute soil

moisture, and facilitate deeper water recharge, contributing to the management of water resources on the watershed scale. The positive and negative effects of intercropped species on hydrological ecosystem services should be tested extensively in other rubber-based agroforestry systems (e.g., rubber with coffee and cocoa) in the future.

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